DAY-1

50 DAY AI CHALLENGE

Python Basics

1. what is python.

Python is a high level interpreted programming language and object-Oriented programming language Created by Guido Van Rossum and first Released in 1991.

2. why python called a high level language.

because its Syntax it close to human language and it hides low level details like memory management and hardware operations. this allows developers to focus on solving problems instead of managing the computer's internal workings.

3. what is an interpreted language

python code is not Directly executed by the CPU.

it follows Execution Flow

python code

↓

Interpreter

↓

Bytecode.

↓

Python Virtual Machine

↓

Machine code

↓

output

Advantage
1. Easy debugging
2. Faster development.

Disadvantage
slightly slower than Compiled clanguage

4. python Compiled or interpreted?
python is both
First, python interpreter Compiles the source code into bytecode (.pyc). then the python Virtual machine executes the bytecode line by line.

5. why python is popular in AI.
python Dominates AI because of its ecosystem

~~library~~.
If we want to do or need a numerical computing we use Numpy library

like this
for pandas - Data Analysis
Matplotlib - Visualization.
scikit-learn :- Machine learning.
pytorch & Tensorflow - Deep learning.
OpenCV - Computer vision
NLTK - Natural language processing.

→ Python Features

1. Easy Syntax
2. Open Source
3. Cross - platform
4. Interpreted
5. Object Oriented
6. Large Community
7. Huge library support
8. Dynamically typed
9. Automatic memory management
10. portable

→ Variables

is named as memory location used to store data.

Ex:- name = "Sushmitha"
age = 20

* python does not require variable declaration
so. python is dynamically typed

Ex:- age = 20.
It automatically determines that this is an integer

→ Data types.

Data types defined the type of value that a variable can store. python automatically identifies the data type based on the assigned value

Data types

	Ex
int → integer	10, -5, 2 . . .
float → Decimal number	8.14, 5.0
complex → Complex number	2+3j
str → Text/String	"Hello"
bool → boolean values	True, False
list → Ordered, mutable collection	[1, 2, 3]
tuple → Ordered, immutable collection	(1, 2, 3)
Set → Unordered Collection of Unique Elements	{1, 2, 3}
dict → key - Value pairs	{"name": "sushmitha", "age": 20}
None Type → It represents no value	None

→ which data types are mutable and immutable in python?

mutable : List, Set, dict

immutable : int, float, complex, str, bool, tuple, Nonetype

mutable	immutable
* it can be changed after they are Created	* it cannot be changed once they are Created
* Memory address usually remains the same after modification	* Memory address changes when the value is reassigned

→ print Function

<u>Syntax</u>:
print (object)

return type of input() is "<u>String</u>"

→ input function
input ()
Ex: ~~tape~~ name = input(—"—)

→ Type Casting
Converting One data type into another

→ 1 Implicit type Casting.
It is also called automatic type conversion.
means it automatically converts One data type into another ~~do~~.

→ 2 Explicit type Casting
~~Doxo~~ is the process of manually converting one data type into another using built in functions Such as ~~like~~ int(), float(), str(), bool(), Etc

→ f - String. (formatted string literal)
is a feature in python used to insert Variables or Expressions directly into a string ~~be~~

<u>Syntax</u>
f"Text {variable}"

→ Memory Management.
It automatically manages memory using reference Counting and a garbage collector, so programmers don't need to manually free memory.